FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Examination of the Newborn

Clinical skill | Week 1 · Breathing, fever & the clinical methods | Dr Nabbale

Every baby gets a quick check at birth and a full head-to-toe examination before discharge (and again at the 3-7-day and 6-week visits). Measure and plot weight, length and head circumference, estimate gestation (New Ballard score), test the primitive reflexes, and screen the eyes (red reflex), hips (Barlow and Ortolani), heart (femoral pulses, oxygen saturation), palate, spine, anus and genitalia, while looking for danger signs that need action today.

Must know before the ward round

1. Timing - quick check at birth (breathing, colour, tone, major malformations), full examination within 24 h and before discharge (UCG - do not discharge a baby under 24 h old), then at 3-7 days and 6 weeks.
2. Normal term values - weight 2.5-4.0 kg, length about 48-53 cm, head circumference about 33-37 cm (plot on WHO/INTERGROWTH charts; microcephaly below -2 SD), RR 30-60/min, HR 100-160/min, axillary temperature 36.5-37.5 C.
3. Danger signs - not feeding well, convulsions, RR 60 or more or below 30, grunting, severe chest indrawing, central cyanosis, temperature above 37.5 or below 35.5 C, floppy/no spontaneous movement, jaundice in the first 24 h or yellow palms/soles.
4. Anterior fontanelle - diamond-shaped, about 1-4 cm, soft and flat; closes by 12-18 months. Posterior fontanelle - small triangle, closes by about 6-8 weeks. Bulging = raised pressure (meningitis); sunken = dehydration.
5. Red reflex - examine both eyes with an ophthalmoscope in a dim room; an absent, dull, white (leukocoria) or asymmetric reflex suggests cataract or retinoblastoma and needs urgent ophthalmology referral.
6. Hips - Barlow (adduct and push posteriorly - dislocates an unstable hip) and Ortolani (abduct and lift - clunk of relocation). Risk factors - breech, family history, female, first baby, oligohydramnios; refer positive tests for ultrasound and orthopaedics.
7. Primitive reflexes - Moro (gone by about 4-6 months), rooting and sucking (about 4 months), palmar grasp (3-6 months), plantar grasp (9-12 months), stepping (about 2 months), ATNR (about 6 months). Asymmetric Moro = brachial plexus palsy or clavicle fracture; absent Moro = severe CNS depression.
8. New Ballard score - 6 neuromuscular + 6 physical signs, total -10 to 50, valid up to about 2 weeks; gestation in weeks = (2 x score + 120) / 5 (score 35 = 38 weeks; 40 = 40 weeks).
9. Femoral pulses - weak or absent = coarctation or other duct-dependent lesion; pulse oximetry screening passes if SpO2 is 95% or more in the right hand or a foot with a difference of 3% or less.
10. Palate - look AND feel (with a torch and tongue depressor) to avoid missing a posterior cleft palate; check nostril patency (choanal atresia - cyanosis that improves with crying).
11. Meconium within 24-48 h and urine within 24 h; delayed meconium (over 48 h) - think Hirschsprung disease, anorectal malformation, meconium ileus.
12. Benign findings to recognise (and record) - Mongolian blue spots (very common in African babies; document to avoid suspicion of abuse), erythema toxicum, milia, Epstein pearls, breast engorgement, white or bloody vaginal discharge, umbilical hernia, postaxial polydactyly.

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In a nutshell

The **newborn examination** is a structured, head-to-toe check that answers three questions:

1. **Is this baby well now?** (danger signs, vital signs, feeding)
2. **Is anything abnormal that needs action or referral?** (malformations, hips, eyes, heart, birth injury, infection risk)
3. **Is the baby the expected size for gestation?** (weight, length, head circumference, Ballard score)

It is done **at birth** (quick check), **within 24 hours and before discharge** (full exam), at the **3-7-day** postnatal visit and at **6 weeks**. UCG 2023 (section 17.1.1) asks every health worker to examine all babies after delivery, before discharge, and whenever a baby is seen in the first week.

Why it matters in Uganda: many babies go home within hours of birth, so this may be the **only full examination** a baby gets. Missing a danger sign (sepsis, jaundice, hypothermia), a cleft palate (feeding failure), an imperforate anus, a dislocated hip or a white pupil (retinoblastoma) has lifelong consequences.

> **CLINICAL PEARL**
> The one thing never to forget: **observe before you touch, and keep the baby warm**. Most of the exam (colour, breathing, posture, movements) is done by looking at a calm baby; examine in a warm room, uncover one part at a time, and finish with the things that make babies cry (hips, Moro, mouth).

Definitions & classification

Normal values for a term newborn

Parameter	Normal range	Notes
Weight	**2.5-4.0 kg**	Below 2.5 kg = LBW; above 4.0 kg = macrosomia (think maternal diabetes). Plot by gestation: SGA below 10th centile, LGA above 90th
Length	About **48-53 cm**	Use a length board (two people)
Head circumference (OFC)	About **33-37 cm** (mean about 35 cm)	Largest of 3 measurements over the occiput and above the eyebrows; microcephaly below **-2 SD** (severe below -3 SD)
Respiratory rate	**30-60/min**	Count for a full minute when calm (UCG); periodic breathing (pauses under 20 s) is normal
Heart rate	**100-160/min**	Sleep can be 90-100; crying up to 180
Temperature	**36.5-37.5 C** axillary	Hypothermia below 36.5 C
SpO2	**95% or more** after the first hours	Pre- and post-ductal difference 3% or less
Capillary refill	Under **3 seconds**	Press on the sternum

Parameter	Normal range	Notes
Weight change	Loss up to **7-10%** in the first 3-5 days; regain birth weight by **10-14 days**	More than 10% = feeding problem
First urine	Within **24 h**	
First meconium	Within **24 h** (almost all term babies by 48 h)	Delay = obstruction until proven otherwise

Gestational age categories

Category	Gestation
Preterm	Below 37 weeks
Term	37+0 to 41+6 weeks
Post-term	42 weeks or more

Equipment

Item	Use
Warm, well-lit room, clean surface or radiant warmer	Prevent hypothermia; see colour and jaundice
Hand-washing facilities / alcohol rub, gloves	Infection prevention
Weighing scale (zeroed, 10-20 g accuracy)	Weight
Non-stretch tape measure	OFC (and length if no board)
Length board (infantometer)	Length
Thermometer (digital, low-reading)	Temperature
Stethoscope (neonatal or small paediatric)	Heart and lungs
Watch with second hand	RR, HR
Pulse oximeter with neonatal probe	SpO2 (right hand and a foot)
Ophthalmoscope	Red reflex
Pen torch and tongue depressor (or clean gloved finger)	Palate, mouth
Growth chart (WHO / INTERGROWTH-21st / Fenton) and **New Ballard chart**	Plotting and gestation
Child Health Card / mother's passport, newborn record	Documentation, immunisation
Glucometer (for at-risk babies)	Hypoglycaemia screening

Indications & contraindications

When to examine

Time	Purpose
At birth (first minute and first hour)	Breathing, HR, colour, tone, obvious malformations; Apgar; skin-to-skin; vitamin K, eye ointment, cord care
Within 24 h and before discharge	Full head-to-toe exam, measurements, Ballard if gestation uncertain, hips, red reflex, palate, anus, spine, femoral pulses, feeding observation

Time	Purpose
3-7 days (postnatal visit)	Jaundice, feeding, weight, cord, eyes, skin, danger signs
6 weeks	Repeat hips, red reflex, heart, testes; growth; development (social smile); immunisation
Any sick visit in the first month	Danger-sign assessment (IMNCI)

Contraindications and cautions

- **Defer a full examination** while a baby is being resuscitated or is unstable: stabilise first (ABC, warmth, glucose).
- **Do not perform Barlow/Ortolani forcefully or repeatedly** (can damage the hip), and **avoid them** in a baby with a suspected limb fracture.
- **Do not force the legs** of a baby born by breech (UCG).
- In **suspected neonatal tetanus**, minimise stimulation (examine after sedation).
- Always obtain the **mother's permission** and explain what you are doing.

Procedure step by step

A. Preparation

1. **Read the records**: maternal age, parity, ANC findings, HIV (last VL), syphilis, HBsAg, blood group, gestation by dates/scan, labour events (ROM, fever, meconium, mode of delivery), resuscitation, Apgar, birth weight, drugs given (vitamin K, eye ointment, NVP).
2. **Introduce yourself**, confirm the baby's identity (name band), explain the exam, get consent; ask the mother about **feeding, urine, stool, and any concerns**.
3. **Wash hands**; warm hands and stethoscope; warm room, no draught.
4. **Observe the baby quietly** in the mother's arms or cot before undressing.

B. General inspection (hands off)

5. **Colour**: pink, pale, **central cyanosis** (tongue/lips), peripheral acrocyanosis (normal in first 24-48 h), **jaundice** (in daylight; blanch skin), plethora.
6. **Breathing**: count **RR for 60 seconds**; look for **grunting, flaring, chest indrawing**, apnoea.
7. **Posture and movement**: term baby lies **flexed** with symmetrical spontaneous movements; note a limp or "frog-leg" posture (hypotonia), asymmetry (palsy, fracture), jitteriness, seizures.
8. **Cry**: normal, weak, high-pitched (CNS), hoarse (hypothyroidism, laryngeal).
9. **Dysmorphic features**: overall appearance (e.g. Down syndrome: upslanting palpebral fissures, flat facial profile, epicanthic folds, single palmar crease, sandal gap, hypotonia).

C. Measurements

10. **Weigh** naked on a zeroed scale; record to the nearest 10-20 g.
11. **Head circumference**: tape over the occiput and just above the eyebrows; take the largest of three.
12. **Length**: two people, baby supine on a length board, legs extended.
13. **Temperature** (axillary, 3-5 min or digital beep).
14. **Plot** all three on the appropriate chart; classify **SGA/AGA/LGA**; note microcephaly/macrocephaly.

D. Head-to-toe examination

15. **Head**: shape, **moulding**, **caput**, **cephalhaematoma**, **subgaleal swelling**; palpate **sutures** (overriding, widely open, craniosynostosis ridge) and **fontanelles** (size, tension: flat, bulging, sunken).
16. **Face**: symmetry at rest and crying (**facial palsy**), dysmorphism.
17. **Eyes**: open spontaneously when the baby is held upright and gently rocked; look for discharge, swelling, subconjunctival haemorrhage (benign), coloboma, cloudy cornea; **red reflex** (see section F).
18. **Ears**: position (top of ear in line with the outer canthus), shape, **pre-auricular tags/pits** (consider hearing screening; renal anomalies if with other features).
19. **Nose**: patency of each nostril (watch misting on a cold surface or listen; choanal atresia → cyanosis at rest that improves with crying).
20. **Mouth**: **inspect the palate with a torch and tongue depressor and palpate with a clean gloved finger** (assessing suck at the same time); cleft lip/palate, **Epstein pearls** (benign), natal teeth, tongue-tie, thrush, micrognathia (Pierre Robin).
21. **Neck**: lumps (cystic hygroma, sternomastoid tumour, goitre), webbing, skin folds; **clavicles** (run fingers along: crepitus, step, swelling = fracture).
22. **Chest**: shape, symmetry, nipples (supernumerary, breast engorgement is normal); **RR, recession**; auscultate both sides (air entry, crackles).
23. **Heart**: apex position (right-sided in dextrocardia/pneumothorax), **HR and rhythm**, heart sounds, **murmurs** (many soft systolic murmurs in the first days are innocent/transitional, but any murmur with cyanosis, poor feeding, tachypnoea or weak pulses is serious); **femoral pulses** (and brachial-femoral comparison); **pulse oximetry** right hand and a foot where available.
24. **Abdomen**: shape (distended or scaphoid: diaphragmatic hernia), **liver** (normal edge up to 2 cm below the costal margin), spleen tip, kidneys (balloting), masses, **umbilical cord** (**2 arteries and 1 vein**; single umbilical artery → look for renal/other anomalies), cord stump (clean, dry, chlorhexidine), hernias (umbilical hernias are common and usually close by 2-5 years).
25. **Groins**: inguinal hernia, femoral pulses (if not already).
26. **Genitalia (male)**: penis length, **meatus position** (hypospadias, epispadias: **do not circumcise**), foreskin, **both testes in the scrotum** (palpate from groin down), hydrocele (transilluminates, usually resolves), scrotal swelling.
27. **Genitalia (female)**: labia, clitoris size, vaginal opening; **white discharge or small withdrawal bleed** is normal (maternal oestrogen). **Ambiguous genitalia** (clitoromegaly, fused labia, impalpable gonads) = possible **congenital adrenal hyperplasia**: do not assign sex, check glucose and electrolytes, urgent senior referral.
28. **Anus**: **patency and position**; ask whether meconium has passed (a fistula can pass meconium yet still be abnormal: look at the position).
29. **Limbs**: count **fingers and toes**, look for **polydactyly** (postaxial extra digit on the little-finger side is common and often familial in African babies), syndactyly, **palmar creases**, **talipes** (positional talipes corrects fully with gentle passive movement; structural club foot does not → refer for Ponseti casting), limb length, movement of each limb, **brachial plexus palsy** (waiter's tip), fractures.
30. **Hips**: Barlow and Ortolani (section G).
31. **Back and spine**: turn the baby prone, supported over your hand (ventral suspension also tests tone); inspect the whole spine for **dimples** (especially above the natal cleft or deep with no visible base), **hair tufts, haemangiomas, lipomas, sinuses**, scoliosis, **open neural tube defects** (myelomeningocele: cover with sterile saline-soaked gauze, nurse prone, refer).
32. **Skin**: jaundice extent, pallor, bruising, petechiae, rashes, pustules; **birthmarks** (Mongolian blue spots: grey-blue patches over the sacrum/buttocks, benign, document them), stork marks, port-wine stain, café-au-lait macules (6 or more = neurofibromatosis), haemangioma (usually appears after birth), **erythema toxicum** (blotchy red with yellow-white centres, day 1-3, baby well), **milia**, **miliaria** (heat rash), peeling (post-term).

E. Neurological examination and primitive reflexes

33. **Alertness and state**: alert, sleepy, irritable, lethargic.
34. **Tone**: posture; **pull-to-sit** (head lag: term babies show some head lag but should not be floppy); **ventral suspension** (term baby holds head briefly in line with the trunk, limbs flexed); scarf sign.
35. **Primitive reflexes** (do Moro last because it makes the baby cry):

Reflex	How to elicit	Normal response	Disappears by	Abnormal if
Rooting	Stroke the cheek near the mouth	Turns head toward the stimulus, opens mouth	About **4 months**	Absent: CNS depression, prematurity
Sucking	Clean finger on the palate	Strong rhythmic suck	About **4 months** (becomes voluntary)	Weak/absent: prematurity, sepsis, HIE, tetanus (trismus)
Palmar grasp	Press a finger into the palm	Fingers flex and grip	**3-6 months**	Absent in one hand: Klumpke palsy; persistent: CP
Plantar grasp	Press the sole below the toes	Toes curl	**9-12 months**	Absent: spinal lesion
Moro (startle)	Support head and back, let the head drop about 1-2 cm suddenly (or pull slightly off the bed and release)	Arms **abduct and extend**, hands open, then **adduct** (embrace) with a cry	About **4-6 months**	**Asymmetric**: **Erb's palsy, clavicle/humerus fracture**; **absent**: severe CNS depression, HIE; **persistent**: CP
Stepping/placing	Hold upright with soles touching the surface	Stepping movements	About **2 months**	
Asymmetric tonic neck reflex (ATNR)	Turn the head to one side (supine)	"Fencing" posture: arm and leg extend on the face side, flex on the occiput side	About **6 months**	Obligatory or persistent: CP
Galant (trunk incurvation)	Stroke along one side of the spine (prone)	Trunk curves toward the stroked side	About **4-6 months**	Absent: spinal cord lesion

F. Red reflex

36. Dim the room; set the ophthalmoscope to **0** dioptres; stand **about 30-50 cm** away; shine the light into both pupils at once (and each eye separately).
37. **Normal**: a **symmetrical red or orange-red glow** in both pupils (it can look darker/orange in babies with more pigmented fundi: compare the two sides).
38. **Abnormal**: **absent or dull reflex**, **white pupil (leukocoria)**, dark spots, or **asymmetry → cataract, retinoblastoma**, glaucoma, vitreous haemorrhage, retinal detachment → **urgent ophthalmology referral** (UK NIPE: seen within 2 weeks; leukocoria same week).

G. Hip examination (DDH)

39. Baby **supine, relaxed** (after a feed is ideal), on a firm surface, nappy removed; examine **one hip at a time**, stabilising the pelvis with the other hand.
40. **Barlow test (provocation)**: hip and knee flexed to 90 degrees, hip **adducted**; apply **gentle downward (posterior) pressure** through the knee. A palpable "clunk" as the femoral head **slips out** of the acetabulum = **dislocatable hip**.
41. **Ortolani test (relocation)**: from the same position, **abduct** the hip while **lifting the greater trochanter forward** with the fingers. A "clunk" as the femoral head **slides back in** = **dislocated, reducible hip**.

42. Look also for **limited abduction**, asymmetric thigh/gluteal skin folds, **leg-length discrepancy** (Galeazzi sign: knees at different heights with hips and knees flexed). Soft "clicks" without movement are usually ligamentous and not significant.
43. **Risk factors** for DDH: **breech** presentation (especially after 36 weeks), **family history** in a first-degree relative, **female**, **first-born**, **oligohydramnios**, large baby, **tight swaddling with hips extended** (a traditional practice in some communities).

H. Gestational age (New Ballard score)

44. Score **6 neuromuscular signs**: posture, square window (wrist), arm recoil, popliteal angle, scarf sign, heel-to-ear.
45. Score **6 physical signs**: skin, lanugo, plantar surface (creases), breast, eye/ear, genitals.
46. Each sign scores **-1 to 5**; total **-10 to 50**. **Gestation (weeks) = (2 × score + 120) ÷ 5.**

Total score	Weeks
-10	20
0	24
10	28
20	32
25	34
30	36
35	38
40	40
45	42
50	44

Best done within **12-20 hours** (valid to about 2 weeks; neuromuscular items are unreliable in sick or sedated babies). Accuracy about ± 2 weeks.

Feature	Preterm (about 32 weeks)	Term
Skin	Thin, red, smooth, visible veins	Thick, pale, cracking/peeling, few vessels
Lanugo	Abundant	Thinning, bald areas
Plantar creases	Faint red marks, anterior transverse crease only	Creases over the whole sole
Breast	Flat areola, no bud or 1-2 mm bud	Full areola, 5-10 mm bud
Ear	Soft, slow recoil	Formed, firm, instant recoil
Genitals (male)	Testes in upper canal, few rugae	Testes down, good rugae
Genitals (female)	Prominent clitoris, small labia minora	Labia majora cover minora and clitoris
Posture	Arms and legs more extended	Fully flexed
Square window	45-60 degrees	0 degrees
Scarf sign	Elbow passes midline	Elbow does not reach midline

I. Finish

47. Dress and wrap the baby, return to the mother (skin-to-skin), **observe a breastfeed** (attachment, effective suckling: UCG).

48. **Explain findings** to the mother in plain words; address concerns.
49. **Document** everything (including normal findings and benign birthmarks) and plan follow-up.
50. **Check routine care**: vitamin K, tetracycline eye ointment, chlorhexidine cord care, **BCG and OPV0** before discharge (hepatitis B birth dose where introduced), HIV prophylaxis (NVP or AZT + NVP) for HIV-exposed babies, risk-based antibiotics, **danger-sign counselling** and the **3-7-day** review.

Clinical acumen

Focused history (what to find out before you touch the baby)

Question	Why it matters
Gestation (LNMP, early scan)	Expected size and maturity; compare with Ballard
Maternal illness: diabetes, hypertension, HIV (VL), syphilis, HBsAg, TB, fever	Hypoglycaemia, SGA, infection risk, prophylaxis
Drugs in pregnancy/labour: anticonvulsants, opioids, MgSO4	Malformations, depressed baby
Liquor volume: oligohydramnios (renal anomalies, DDH, lung hypoplasia), polyhydramnios (oesophageal atresia, GI obstruction)	Directs the exam
Labour: ROM duration, maternal fever, meconium, breech, instrumental delivery, shoulder dystocia	Sepsis risk, DDH, birth injury
Resuscitation and Apgar	HIE risk
Family history: DDH, cleft, heart disease, deafness, sickle cell, early childhood eye problems	Targeted screening
Feeding since birth, urine and meconium passed	Obstruction, feeding problems
Mother's concerns	Mothers often notice the problem first

Examination: what matters most

Danger signs, vital signs and measurements first; then the "**easy-to-miss five**": **palate**, **red reflex**, **hips**, **femoral pulses**, **anus/spine**.

Red flags

RED FLAG

- **Any IMNCI danger sign**: not feeding, convulsions, RR 60 or more or below 30, grunting, severe indrawing, central cyanosis, temperature above 37.5 or below 35.5 C, floppy or no movement
- **Jaundice in the first 24 h**, or yellow palms/soles
- **Absent or white red reflex**
- **Absent femoral pulses**, SpO2 below 95% or a difference above 3% between hand and foot, cyanosis, a loud or harsh murmur
- **Bilious vomiting**, abdominal distension, **no meconium by 48 h**, imperforate anus
- **Bulging or very large fontanelle**, OFC above +2 SD or below -2 SD
- **Ambiguous genitalia** (possible salt-losing CAH)
- **Open neural tube defect**, gastroschisis/omphalocele
- **Excessive drooling and choking with feeds** (oesophageal atresia: pass an orogastric tube)
- **Weight loss above 10%** or a mother worried that the baby "is not right"

Thinking like a clinician

- **A "normal" exam in a baby who is not feeding is not normal.** Feeding problems are the commonest early sign of sepsis, hypoglycaemia, HIE and malformations.
- **Use the history to target the exam**: polyhydramnios → pass a tube to exclude oesophageal atresia; breech → careful hip exam; diabetic mother → glucose, heart, sacrum (caudal regression); instrumental delivery → scalp and facial nerve.
- **Peripheral cyanosis (hands and feet) is normal in the first days; central cyanosis (tongue) is never normal.**
- **Distinguish benign from serious skin findings**: Mongolian spots vs bruises (spots are present at birth, uniform, non-tender, do not change quickly); erythema toxicum vs pustular infection (well baby, centres sterile, changes location).
- **One anomaly → look for more** (e.g. single umbilical artery → kidneys; ear tags → hearing and kidneys; Down syndrome features → heart (AVSD), duodenal atresia, hypothyroidism).
- **Re-examine at 6 weeks**: hips and testes can change, cataracts may become visible, murmurs appear as pulmonary pressure falls.

Clinical workup

For a skill topic, the "workup" is the pre-examination preparation and the post-examination checks.

Step 0: First 5 minutes (at birth or on receiving a baby)

1. **Breathing or crying?** If not, start HBB resuscitation (see Neonatal Resuscitation).
2. **Dry, warm, skin-to-skin**, cover the head.
3. **Quick look**: colour, tone, obvious malformation (cleft, NTD, abdominal wall defect, limb anomaly, ambiguous genitalia).
4. **Danger signs** present? → act (oxygen, glucose, antibiotics, SCBU).
5. **Breastfeed within 1 hour**.
6. **Vitamin K 1 mg IM** (0.5 mg preterm/LBW per local protocol), **tetracycline eye ointment**, **chlorhexidine** to the cord, identification band.

Step 1: Focused history

- [ ] Maternal records reviewed (HIV/VL, syphilis, HBsAg, blood group, diabetes, hypertension, fever).
- [ ] Gestation by best dates.
- [ ] Liquor, labour, delivery mode, resuscitation, Apgar.
- [ ] Feeds, urine, meconium since birth.
- [ ] Family history; mother's concerns.

Step 2: Focused examination

- [ ] Vitals and measurements, plotted.
- [ ] Head-to-toe with the "easy-to-miss five" (palate, red reflex, hips, femorals, anus/spine).
- [ ] Neuro: tone, alertness, reflexes.
- [ ] Ballard if gestation uncertain.
- [ ] Observe a breastfeed.

Step 3: Bedside tests

Test	Indication	Normal / action
Glucose	At risk (preterm, SGA, LGA/IDM, asphyxia, sepsis, hypothermia, poor feeding)	Treat below **2.6 mmol/L**
Pulse oximetry	All babies where available (CCHD screening), and any with cyanosis/distress	**Pass**: 95% or more in right hand or foot with difference 3% or less; **fail**: below 90%; 90-94% or difference above 3% → repeat once, then echo/senior review
Transcutaneous bilirubin	Visible jaundice after 24 h	Plot; serum if high
Orogastric tube	Polyhydramnios, frothy secretions, choking	Tube coils in the oesophagus (on X-ray) = oesophageal atresia
Transillumination	Scrotal swelling, suspected pneumothorax	Hydrocele transilluminates

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management	Availability
Hip ultrasound	Positive Barlow/Ortolani, or risk factors (breech, family history)	Dysplasia/dislocation (Graf classification)	Pavlik harness; orthopaedic follow-up	Regional/national referral, private
Echocardiography	Murmur with symptoms, failed oximetry, weak femorals, Down syndrome	Structural heart disease	Prostaglandin/surgery referral	National referral (Uganda Heart Institute), some regional
Renal ultrasound	Single umbilical artery with other anomalies, ear tags with other features, abdominal mass	Renal anomalies	Urology/nephrology	Most hospitals
Spinal ultrasound	Atypical sacral dimple, tuft, haemangioma	Tethered cord	Neurosurgery (CURE Hospital Mbale, Mulago)	Referral hospitals

Test	When	Expected / abnormal finding	How it changes management	Availability
Karyotype/genetics	Dysmorphism (e.g. Down syndrome), ambiguous genitalia	Trisomy 21, DSD	Counselling, CAH management	Limited (Kampala/private)
Glucose, electrolytes, 17-OH-progesterone	Ambiguous genitalia	Salt-wasting CAH	Hydrocortisone, fluids	Limited
Hearing screen (OAE)	Risk factors (meningitis, severe jaundice, gentamicin, ear anomalies, family history)	Refer result	Audiology	Few centres
Thyroid function	Prolonged jaundice, Down syndrome, features of hypothyroidism	Low T4, high TSH	Levothyroxine	Referral hospitals

Step 5: Putting it together

Model summary line (normal): "A **20-hour-old** term (39 weeks by early scan; Ballard 38) male, birth weight **3.25 kg (AGA)**, OFC 34.5 cm, length 50 cm, born by SVD after an uncomplicated labour, mother HIV-negative, RPR negative, HBsAg negative. He is breastfeeding 8 times in 24 h with good attachment, has passed urine and meconium. Examination is normal: RR 44, HR 136, T 36.9 C, SpO2 98% right hand and 97% foot, palate intact, red reflexes present and symmetrical, hips stable, femoral pulses normal, anus patent, spine intact, Mongolian spots over the sacrum documented. Vitamin K, eye ointment, chlorhexidine, BCG and OPV0 given. Plan: discharge after 24 h with danger-sign counselling and 3-7-day review."

Worked example (Ballard): a baby whose LNMP is unknown scores **25** → (2 × 25 + 120) ÷ 5 = **34 weeks**. If she weighs 1.6 kg, plot at 34 weeks: about the 10th centile → borderline SGA; she needs KMC, feeding support and glucose monitoring.

Management

The "management" of the newborn examination is **what you do with what you find**.

1. Abnormal findings: action table

Finding	Likely meaning	Action
Any **danger sign**	Sepsis, hypoglycaemia, hypothermia, HIE, CHD	Stabilise; **ampicillin + gentamicin**; glucose; warmth; SCBU/referral
Jaundice day 1 / palms and soles yellow	Haemolysis, sepsis	**TSB urgently**, phototherapy per thresholds
Weight below 2.5 kg	LBW	KMC, feeding support; **below 1.5 kg or below 32 weeks: refer** (UCG)
Absent/abnormal red reflex	Cataract, retinoblastoma	**Urgent ophthalmology referral** (e.g. Mulago, Ruharo, Mengo eye units)
Positive Barlow/Ortolani, limited abduction	DDH	Ultrasound and **orthopaedic referral** (CoRSU, Mulago, regional ortho); avoid tight swaddling; Pavlik harness if confirmed
Breech or family history, normal exam	DDH risk	Hip ultrasound at about 6 weeks (where available) and careful 6-week exam
Club foot (structural)	Congenital talipes equinovarus	Refer early for **Ponseti casting** (clubfoot clinics at CoRSU, Mulago and many regional referral hospitals)

Finding	Likely meaning	Action
Positional talipes	Intrauterine moulding	Reassure; passive stretching
Cleft lip/palate	Feeding problem	Support breastfeeding or **cup/spoon EBM** (UCG); refer to cleft surgery (CoRSU, Mulago); check for other anomalies
Absent femoral pulses / failed oximetry / cyanosis	Duct-dependent CHD, coarctation	Senior review, oxygen cautiously, **echo**, refer (Uganda Heart Institute)
Imperforate anus / no meconium by 48 h / bilious vomiting	Anorectal malformation, Hirschsprung, malrotation	**Nil by mouth**, NGT on free drainage, IV fluids, urgent surgical referral
Undescended testis	Cryptorchidism	Recheck at 6-8 weeks; refer for orchidopexy by **6-12 months** if not descended (bilateral impalpable with hypospadias = DSD, urgent)
Hypospadias	Urethral meatus on the underside	**Do not circumcise**; refer to paediatric surgery
Ambiguous genitalia	CAH, DSD	Do not assign sex; glucose, electrolytes; urgent referral
Open neural tube defect	Myelomeningocele	Cover with sterile saline-soaked gauze, nurse prone, avoid latex, **refer** (CURE Children's Hospital Mbale); folic acid counselling for the next pregnancy
Sacral dimple (deep, above natal cleft, with tuft/haemangioma)	Occult spinal dysraphism	Spinal ultrasound; neurosurgery
Gastroschisis/omphalocele	Abdominal wall defect	Cover with cling film/sterile bag, keep warm, NGT, IV fluids, urgent surgical referral
Down syndrome features	Trisomy 21	Sensitive disclosure with both parents; echo; thyroid; feeding support; refer
Asymmetric Moro / limp arm	Erb's palsy, clavicle fracture	See Common Neonatal Disorders (birth injuries)
Large tense fontanelle, fast-growing OFC	Hydrocephalus, meningitis	Senior review; cranial US; LP if infection suspected
Polydactyly (postaxial, narrow stalk)	Often familial, isolated	Refer for removal; avoid tying off at home with thread unless your unit's protocol supports it
Natal teeth (loose)	Aspiration risk	Dental review
Tongue-tie with feeding problems	Ankyloglossia	Assess feeding; frenotomy if feeding affected

2. Benign findings: reassure and document

Finding	Explanation to the mother
Mongolian blue spots	Normal pigment, very common in African babies, fade over years; **record them**
Erythema toxicum	Harmless rash in the first days; comes and goes
Milia	Tiny white spots on the nose; disappear
Epstein pearls	White cysts on the palate; disappear
Breast engorgement / "witch's milk"	Mother's hormones; **do not squeeze** (abscess risk)
Vaginal discharge/small bleed	Mother's hormones; settles in days
Subconjunctival haemorrhage	From birth pressure; clears in 1-2 weeks

Finding	Explanation to the mother
Caput, moulding	Settle in 1-3 days
Umbilical hernia	Common; usually closes by 2-5 years; no coins or binders
Acrocyanosis	Blue hands and feet in the first days; keep warm
Hydrocele	Fluid around the testis; usually resolves by 1 year

3. Communication and documentation

- Use plain, honest language; deliver unexpected findings (e.g. Down syndrome, cleft) **with both parents present** where possible, in private, with a senior.
- Do not blame the mother; explain the next steps and referral clearly; write a referral letter.
- Document **measurements with centiles**, **all systems** including normal findings, and **benign marks**; record vaccines and prophylaxis given.

4. Discharge checklist (UCG)

- **At least 24 h old**, feeding well (8 or more feeds/24 h), temperature normal, no danger signs.
- Do **not** plan early discharge for **small (LBW/preterm)** babies or babies not feeding well.
- **BCG and OPV0** given; HIV prophylaxis started if exposed; vitamin K and eye ointment done.
- Counsel on **danger signs** (refusal to feed, excessive crying, cord bleeding, fever, bulging fontanelle, abdominal distension, grunting, yellow palms/soles), **cord care**, **warmth**, **exclusive breastfeeding**, **sleeping under a treated net**.
- **Next visit at 3-7 days**; immunisation at **6 weeks**.

OSCE checklist

Step	Done
Introduces self, confirms identity, explains, gains consent; asks about feeding, urine, meconium, concerns	☐
Washes/warms hands; warm room; reviews maternal and birth records	☐
Observes before touching: colour, breathing (counts RR 60 s), posture, movement, cry	☐
Measures weight, OFC, length, temperature; plots on chart; classifies SGA/AGA/LGA	☐
Head: fontanelles, sutures, swellings (caput/cephalhaematoma/subgaleal)	☐
Face, eyes (red reflex both eyes), ears, nose patency	☐
Mouth: inspects **and** palpates palate; assesses suck	☐
Neck and clavicles	☐
Chest: RR, recession, auscultation	☐
Heart: HR, sounds, murmurs, **femoral pulses**, SpO2 (hand and foot)	☐
Abdomen: liver, spleen, masses, cord vessels and stump	☐
Genitalia (testes/meatus or female genitalia) and **anal patency**	☐
Limbs: digits, palmar creases, feet (talipes), movement	☐
Hips: Barlow and Ortolani, gently, one side at a time	☐
Back and spine in ventral suspension (tone, dimples, NTD)	☐
Skin: jaundice, birthmarks, rashes	☐

Step	Done
Neuro: tone, alertness, rooting, sucking, grasp, Moro (last)	☐
Ballard score if gestation uncertain	☐
Re-dresses baby, returns to mother, observes a breastfeed	☐
Explains findings; counsels on danger signs, cord care, warmth, feeding, vaccines, follow-up	☐
Documents and presents a concise summary with a plan	☐

Complications

The examination itself is safe if done gently; risks and pitfalls include:

- **Hypothermia** from a long exam with the baby uncovered.
- **Hip injury** from forceful or repeated Barlow/Ortolani.
- **Infection** from unwashed hands or equipment.
- **Missed diagnoses**: posterior cleft palate (looked but did not feel), cataract/retinoblastoma (no red reflex check), coarctation (femorals not felt), imperforate anus (not inspected), DDH (hips not examined), sepsis (danger signs not checked).
- **False reassurance** when a mother's concern is dismissed.
- **Mislabelling** of Mongolian spots as bruises (child-protection harm) when not documented.

Ward-round & exam pearls

Model presentation

"**Baby boy of Nabirye**, **18 hours old**, born at **38 weeks** by early scan (Ballard 36-38) by SVD after a 9-hour labour, ROM 4 hours, clear liquor, **Apgar 8 and 10**; no resuscitation. Birth weight **3.1 kg (50th centile, AGA)**, OFC **34 cm (25th centile)**, length **49 cm**.

Mother is a 24-year-old para 2, **HIV-positive on TLD with a suppressed viral load (below 50 copies) at 34 weeks**, RPR negative, HBsAg negative, blood group O positive. The baby is **low-risk HIV-exposed** and received **NVP 15 mg** at 2 hours; plan **NVP daily for 6 weeks** and **DNA PCR at 6 weeks**. He is **exclusively breastfeeding**, 7 feeds so far, good attachment; passed urine and meconium.

On examination he is pink, active, with a normal cry; RR 46, HR 140, T 36.8 C, SpO2 98% hand and 97% foot. Anterior fontanelle 2 × 2 cm, flat. Palate intact on inspection and palpation. **Red reflexes present and symmetrical.** Heart sounds normal, no murmur, **femoral pulses normal**. Abdomen soft, 3 cord vessels, stump clean with chlorhexidine. Both testes descended, meatus normal, anus patent. **Hips stable** on Barlow and Ortolani. Spine intact. **Mongolian spots** over the sacrum, documented. Normal tone; rooting, suck, grasp and symmetrical Moro present.

Vitamin K, tetracycline eye ointment, BCG and OPV0 given. **Assessment:** healthy term AGA HIV-exposed infant. **Plan:** discharge after 24 h, counsel on danger signs and exclusive breastfeeding, NVP adherence, review at 3-7 days, 6-week immunisation and DNA PCR."

Viva questions (with answers)

1. **What is the normal RR and HR of a newborn?** RR 30-60/min; HR 100-160/min.
2. **How do you perform the Ortolani test, and what does a positive test mean?** Flex hip and knee to 90 degrees, abduct while lifting the greater trochanter; a clunk as the head relocates = a dislocated but reducible hip.

3. **Why examine the red reflex?** To detect cataract and retinoblastoma early; absent or white reflex needs urgent referral.
4. **When does the Moro reflex disappear and what does an asymmetric Moro suggest?** About 4-6 months; brachial plexus palsy or clavicle/humerus fracture.
5. **How do you calculate gestation from the New Ballard score?** (2 × score + 120) ÷ 5.
6. **Why feel the femoral pulses?** To detect coarctation of the aorta or other left-sided obstructive lesions.
7. **Name five benign findings in a newborn.** Mongolian spots, erythema toxicum, milia, Epstein pearls, breast engorgement (also vaginal discharge, subconjunctival haemorrhage, umbilical hernia).

OSCE tips

- Talk to the mother throughout; keep the baby warm; wash hands at the start and end.
- Do the **hips and Moro last** (they make the baby cry).
- **State normal findings aloud** ("palate intact on inspection and palpation; red reflexes present bilaterally").
- End with a **one-line summary and plan**.

Mnemonics & memory aids

Order of the exam: "Look, Measure, Head-to-Toe, Turn, Test": look (colour, breathing, posture), measure (weight, OFC, length, temperature), head-to-toe (supine), turn (back and spine), test (reflexes, hips last).

Easy-to-miss checks: "Please Remember Five Hidden Abnormalities, Seriously": **P**alate, **R**ed reflex, **F**emoral pulses, **H**ips, **A**nus, **S**pine.

Reflex disappearance: "2-4-6-12": stepping by **2** months, rooting/sucking by **4**, Moro/ATNR by about **6**, plantar grasp by **12** months.

Barlow vs Ortolani: "Barlow = Bye-bye (out), Ortolani = Only back In".

High-yield summary

Numbers & doses to know

Item	Value
Normal term weight	**2.5-4.0 kg**
Normal OFC	About **33-37 cm** (mean about 35)
Normal length	About **48-53 cm**
RR / HR	**30-60** / **100-160** per minute
Temperature	**36.5-37.5 C** axillary
SpO2 screening	Pass **95% or more** with hand-foot difference **3% or less**; fail **below 90%**
Weight loss	Up to **7-10%**; regain by **10-14 days**
Urine / meconium	Within **24 h** / within **24-48 h**
Anterior fontanelle closure	**12-18 months**
Posterior fontanelle closure	**6-8 weeks**
Ballard formula	**(2 × score + 120) ÷ 5**
Moro disappears	**4-6 months**

Item	Value
Palmar / plantar grasp	**3-6 months** / **9-12 months**
Discharge	Not before **24 h** (UCG); review at **3-7 days**; **6-week** check
Vitamin K	**1 mg IM** (0.5 mg preterm/LBW)
Undescended testis	Refer for surgery by **6-12 months**

Classic exam traps

- **"Palate looks fine on inspection, so it is intact."** → You must also **palpate** (and see the back with a torch): posterior/submucous clefts are missed.
- **"Bluish patches on the buttocks: suspect abuse."** → Usually **Mongolian spots**: present at birth, benign; **document** them.
- **"Blue hands and feet at 6 hours: cyanotic heart disease."** → **Acrocyanosis** is normal; check the **tongue** for central cyanosis and do oximetry.
- **"A click on hip examination = DDH."** → Soft ligamentous clicks are common; the significant finding is a **clunk** of dislocation/relocation or limited abduction.
- **"Absent red reflex can wait for the 6-week check."** → Wrong. **Refer urgently** (retinoblastoma, cataract).
- **"He passed meconium, so the anus is normal."** → A fistula can pass meconium; **inspect the anal position and patency**.
- **"Circumcise to correct hypospadias."** → Never circumcise: the foreskin is needed for repair.
- **"Squeeze the swollen breasts to remove the milk."** → Wrong: causes mastitis/abscess.

Questions seniors ask

1. **"Is the baby the right size for gestation?"** Give weight, OFC and length with centiles and the Ballard estimate.
2. **"Did you check the palate, red reflex, hips, femorals and anus?"** Say yes, and how.
3. **"What is the mother's HIV status and what prophylaxis has the baby received?"** VL, risk category, NVP ± AZT dose, PCR plan.
4. **"Has the baby fed, passed urine and meconium?"** Times and number of feeds.
5. **"What benign findings did you note?"** e.g. Mongolian spots, erythema toxicum, documented.
6. **"Can this baby go home today?"** Over 24 h, feeding well, normal exam, vaccines and prophylaxis given, mother counselled, follow-up booked.
7. **"What would make you refer?"** Danger signs, absent red reflex, unstable hip, absent femorals/failed oximetry, imperforate anus, ambiguous genitalia, NTD, cleft with feeding failure.

SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	When should a newborn be examined?	Quick check **at birth**; full head-to-toe exam **within 24 h and before discharge** (UCG: do not discharge under 24 h); again at **3-7 days** and **6 weeks**, and at any sick visit.
2	Normal respiratory rate, heart rate and temperature of a term newborn.	RR **30-60/min** (count a full minute), HR **100-160/min**, axillary temperature **36.5-37.5 C**.
3	Normal term weight, length and head circumference.	Weight **2.5-4.0 kg**, length about **48-53 cm**, OFC about **33-37 cm** (mean about 35). Plot on WHO/INTERGROWTH charts.
4	How do you measure head circumference?	Non-stretch tape around the **occiput and just above the eyebrows**; take the **largest of three** measurements; plot it (microcephaly below -2 SD).
5	Anterior and posterior fontanelles: size and closure.	Anterior: diamond, about **1-4 cm**, closes by **12-18 months**. Posterior: small triangle, closes by **6-8 weeks**. Bulging = raised ICP (meningitis); sunken = dehydration.
6	How to check the red reflex.	Dim room, ophthalmoscope at 0 dioptres, **30-50 cm** away, light into both pupils: normal = **symmetrical red/orange glow**. Absent, dull, white or asymmetric → **urgent ophthalmology** (cataract, retinoblastoma).
7	Barlow test.	Hip and knee flexed 90 degrees, hip **adducted**, gentle **posterior pressure** through the knee: a clunk as the head **slips out** = dislocatable (unstable) hip.
8	Ortolani test.	From flexion, **abduct** the hip while **lifting the greater trochanter**: a clunk as the head **slides back in** = dislocated, reducible hip.
9	Risk factors for developmental dysplasia of the hip.	**Breech** presentation, **family history**, **female**, **first-born**, **oligohydramnios**, large baby, **tight swaddling with hips extended**.
10	Moro reflex: how to elicit, normal response, disappearance.	Support head and back, let the head drop 1-2 cm: arms **abduct and extend**, hands open, then **adduct** with a cry. Disappears by about **4-6 months**.
11	Meaning of an asymmetric or absent Moro reflex.	**Asymmetric**: Erb's palsy, clavicle or humerus fracture. **Absent**: severe CNS depression (HIE, sepsis, drugs). **Persistent beyond 6 months**: cerebral palsy.
12	Age of disappearance of primitive reflexes.	Stepping **~2 months**; rooting and sucking **~4 months**; palmar grasp **3-6 months**; Moro and ATNR **~4-6 months**; plantar grasp **9-12 months**.
13	New Ballard score: components and formula.	**6 neuromuscular** (posture, square window, arm recoil, popliteal angle, scarf sign, heel-to-ear) + **6 physical** (skin, lanugo, plantar surface, breast, eye/ear, genitals). Total -10 to 50. **GA = (2 × score + 120) ÷ 5.**
14	Ballard score of 30 equals what gestation?	(2 × 30 + 120) ÷ 5 = **36 weeks**.
15	Pulse oximetry screening for critical congenital heart disease.	Right hand and a foot after 24 h: **pass** if **95% or more** in either with a difference of **3% or less**; **fail** if below 90%; 90-94% or difference above 3% → repeat once, then senior review/echo.
16	Why must you feel the femoral pulses?	Weak or absent femorals (or brachio-femoral delay) suggest **coarctation of the aorta** or another duct-dependent left heart lesion.

#	Question	Answer
17	How to examine the palate.	**Inspect with a torch and tongue depressor AND palpate** with a clean gloved finger (also tests suck). Inspection alone misses posterior/submucous clefts.
18	When should urine and meconium be passed?	Urine within **24 h**; meconium within **24 h** (almost all term babies by 48 h). Delay → Hirschsprung disease, anorectal malformation, meconium ileus.
19	Normal weight loss after birth.	Up to **7-10%** in the first 3-5 days; regain birth weight by **10-14 days**. More than 10% → feeding problem.
20	List benign newborn findings.	**Mongolian blue spots**, **erythema toxicum**, **milia**, **Epstein pearls**, breast engorgement, white/bloody vaginal discharge, subconjunctival haemorrhage, acrocyanosis, umbilical hernia, hydrocele.
21	Why document Mongolian spots?	They are grey-blue patches (sacrum/buttocks), very common in African babies, present at birth and benign; documenting them prevents later **confusion with bruising/abuse**.
22	Positional vs structural talipes.	**Positional** talipes corrects **fully with gentle passive movement** (reassure, stretch). **Structural club foot** is rigid → refer early for **Ponseti casting**.
23	Action for ambiguous genitalia.	**Do not assign sex**; check **glucose and electrolytes** (salt-wasting **CAH** can cause shock in week 1-2); urgent senior/endocrine referral; sensitive counselling.
24	Management of an open neural tube defect at birth.	Cover with **sterile saline-soaked gauze**, nurse **prone**, keep warm, avoid latex, IV access if needed, **refer** to neurosurgery (e.g. CURE Children's Hospital Mbale); folic acid counselling for future pregnancies.
25	Single umbilical artery: significance.	Normal cord has **2 arteries + 1 vein**. A single artery is associated with **renal and other anomalies**: examine carefully; renal ultrasound if other findings.
26	Central vs peripheral cyanosis in the newborn.	**Acrocyanosis** (blue hands/feet) is normal in the first 24-48 h. **Central cyanosis** (blue tongue/lips) is never normal: check SpO2, think heart/lung disease.
27	Routine care to confirm at the newborn examination (Uganda).	**Vitamin K 1 mg IM** (0.5 mg preterm/LBW), **tetracycline eye ointment**, **chlorhexidine** cord care, **BCG + OPV0** (HepB birth dose where introduced), HIV prophylaxis if exposed, breastfeeding within 1 h, danger-sign counselling.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. During a discharge examination of a 20-hour-old term baby, you shine an ophthalmoscope into both eyes from 40 cm. The right pupil glows orange-red; the left pupil looks white. The baby is feeding well. What is the MOST appropriate action?

A Reassure; recheck at the 6-week visit
B Prescribe chloramphenicol eye ointment
C Recheck after 1 week in brighter light
D Discharge and advise the mother to watch for squint
E Refer urgently to an ophthalmologist for possible cataract or retinoblastoma

2. A 2-day-old baby girl born by breech CS is examined. With the hip and knee flexed and the hip adducted, gentle backward pressure on the knee produces a clunk as the femoral head slips out of the socket. What is the name of this test and the MOST appropriate next step?

A Ortolani test; reassure
B Galeazzi test; X-ray of the pelvis today
C Barlow test (positive); refer for hip ultrasound and orthopaedic review, and avoid tight swaddling
D Trendelenburg test; physiotherapy
E Barlow test; repeat it 10 times to confirm

3. A mother is worried about grey-blue patches over her 1-day-old son's lower back and buttocks. They were present at birth, are flat, non-tender and uniform in colour. What is the MOST appropriate response?

A Explain these are Mongolian blue spots, a common benign birthmark, and document them in the notes
B Suspect non-accidental injury and inform the police
C Order a coagulation screen and platelet count
D Apply hydrocortisone cream
E Refer to dermatology for laser treatment

4. A term baby is examined at 12 hours. His hands and feet are bluish but his tongue and lips are pink. RR 48, HR 138, temperature 36.8 C, SpO2 98% in the right hand. What is the MOST likely explanation?

A Cyanotic congenital heart disease
B Methaemoglobinaemia
C Persistent pulmonary hypertension
D Acrocyanosis, a normal finding in the first 24-48 hours
E Septic shock

5. A baby's gestational age is unknown. The New Ballard score is 20. What is the estimated gestation and the MOST appropriate implication?

A 28 weeks: very preterm, ventilate electively
B 32 weeks: preterm; needs special newborn care (KMC, feeding support, and referral if very small)
C 36 weeks: late preterm; routine care
D 38 weeks: term; routine care
E 40 weeks: term; discharge at 6 hours

6. A 30-hour-old term baby has a soft systolic murmur, is feeding well and pink. Femoral pulses are hard to feel compared with the brachial pulses. SpO2 is 98% in the right hand and 91% in the foot. What is the MOST appropriate next step?

A Reassure: murmurs in the first days are innocent
B Discharge and review at 6 weeks
C Urgent senior review, echocardiography and referral for possible coarctation/duct-dependent lesion
D Start oral amoxicillin
E Give furosemide and discharge

7. A 1-day-old baby with polyhydramnios on the antenatal scan is drooling frothy saliva and coughs and turns blue with the first breastfeed. What is the MOST appropriate next step?

A Encourage more frequent breastfeeds
B Give gripe water
C Discharge with advice to burp the baby
D Start antibiotics only
E Stop oral feeds, pass an orogastric tube and request an X-ray; keep upright with suction and refer to paediatric surgery

8. You are examining a 1-day-old baby's mouth. On inspection with a torch the palate looks intact, but your gloved finger feels a notch at the back of the hard palate. The baby has had difficulty breastfeeding with milk coming from the nose. What is the MOST appropriate management?

A Reassure; the palate is intact on inspection
B Diagnose a posterior cleft palate; support feeding with expressed breast milk by cup/spoon and refer to a cleft surgery service
C Start bottle feeding with a standard teat
D Give nystatin drops
E Advise the mother to stop breastfeeding and use formula through a nasogastric tube permanently

9. A 36-hour-old baby has not passed meconium. Abdomen is distended and he vomited green fluid twice. What is the MOST appropriate immediate step?

A Give a glycerine suppository and continue breastfeeding
B Reassure; some babies take 72 hours
C Give oral laxatives
D Stop feeds, pass an NGT on free drainage, start IV fluids, inspect the anus, and refer urgently to paediatric surgery
E Give an enema with soap water

10. A baby born after shoulder dystocia has an asymmetric Moro reflex: the right arm does not abduct. There is crepitus and swelling over the right clavicle; the grasp reflex is normal. What is the MOST likely diagnosis?

A Fractured right clavicle
B Cerebral palsy
C Hypoxic-ischaemic encephalopathy
D Klumpke palsy
E Septic arthritis of the shoulder

11. A mother asks when she can take her healthy term baby home. The baby is 10 hours old, breastfeeding well, has passed urine, and the examination is normal. According to UCG 2023, what is the BEST answer?

A Immediately, since the examination is normal

B After the baby is at least 24 hours old, feeding well, with BCG and OPV0 given and danger-sign counselling done

C After 7 days

D Only after the baby regains birth weight

E Only after the cord falls off

12. Which reflex normally persists the LONGEST in infancy?

A Stepping reflex

B Rooting reflex

C Moro reflex

D Plantar grasp reflex

E Palmar grasp reflex

13. A term baby's examination shows clitoromegaly, fused labioscrotal folds and no palpable gonads. The parents ask whether the baby is a boy or a girl. What is the MOST appropriate response?

A Avoid assigning sex, explain that tests are needed, check glucose and electrolytes, and refer urgently for possible congenital adrenal hyperplasia

B Register the baby as a girl because there are no testes

C Register the baby as a boy because of the enlarged phallus

D Reassure and review at 6 weeks

E Arrange surgery in the first week

14. Which examination finding in a newborn MOST needs action before discharge?

A Milia on the nose

B White vaginal discharge in a female baby

C Bilateral breast swelling

D Erythema toxicum on the trunk

E Jaundice visible at 16 hours of age

15. A term baby has a 1 cm deep dimple above the natal cleft with a tuft of hair over it. Neurological examination of the legs is normal and the anus is patent. What is the MOST appropriate next step?

A Reassure: sacral dimples are always benign

B Probe the dimple with a cotton bud

C Arrange a spinal ultrasound and neurosurgical review for possible occult spinal dysraphism

D Shave the tuft and apply antiseptic

E Start antibiotics

Answers & explanations

> **1. E** · Refer urgently to an ophthalmologist for possible cataract or retinoblastoma
>
> E is right: an absent or white red reflex (leukocoria) suggests cataract or retinoblastoma; both need urgent ophthalmology assessment because early treatment preserves vision (cataract) and life (retinoblastoma). A, C and D are wrong: waiting delays sight- and life-saving treatment. B is wrong: there is no conjunctivitis; antibiotic ointment does not treat leukocoria.

2. C · Barlow test (positive); refer for hip ultrasound and orthopaedic review, and avoid tight swaddling

C is right: adduction with posterior pressure that dislocates the hip is a positive Barlow test; with risk factors (breech, female) she needs ultrasound and orthopaedic follow-up (Pavlik harness if confirmed), and families should avoid swaddling with the legs straight. A is wrong: Ortolani relocates a dislocated hip by abduction and lifting. B is wrong: Galeazzi compares knee heights, and X-rays are not useful before the femoral head ossifies (about 4-6 months). D is wrong: Trendelenburg is a walking-child test. E is wrong: repeated forceful testing can damage the hip.

3. A · Explain these are Mongolian blue spots, a common benign birthmark, and document them in the notes

A is right: Mongolian spots (dermal melanocytosis) are very common in African babies, present at birth and benign; documenting them prevents later confusion with bruising. B is wrong: findings present at birth that are uniform and non-tender are not bruises. C is wrong: there is no bleeding problem. D is wrong: steroids do nothing for pigment. E is wrong: they fade over years and need no treatment.

4. D · Acrocyanosis, a normal finding in the first 24-48 hours

D is right: blue hands and feet with a pink tongue and normal SpO2 is peripheral cyanosis (acrocyanosis), normal in the first days; keep warm. A and C are wrong: they cause central cyanosis and low SpO2. B is wrong: methaemoglobinaemia gives central cyanosis (chocolate-coloured blood) with an oximetry reading in the mid-80s. E is wrong: the baby is well with normal vital signs.

5. B · 32 weeks: preterm; needs special newborn care (KMC, feeding support, and referral if very small)

B is right: GA = (2 × 20 + 120) ÷ 5 = 32 weeks. A 32-week baby is very preterm/moderately preterm with risks of hypothermia, poor suck, apnoea and RDS; he needs KMC, feeding support and, per UCG, referral if below 32 weeks or below 1.5 kg. A is wrong: 28 weeks corresponds to a score of 10, and elective ventilation is not indicated. C, D and E are wrong: they correspond to scores of 30, 35 and 40.

6. C · Urgent senior review, echocardiography and referral for possible coarctation/duct-dependent lesion

C is right: weak femoral pulses with a pre-/post-ductal SpO2 difference of 7% (above 3%) and a post-ductal value below 95% suggest a duct-dependent left heart lesion such as coarctation; the baby can collapse when the duct closes. Urgent echo and referral (Uganda Heart Institute) are needed. A and B are wrong: the femoral and oximetry findings make an innocent murmur unlikely. D is wrong: there is no infection. E is wrong: diuretics do not treat duct-dependent disease, and discharge is unsafe.

7. E · Stop oral feeds, pass an orogastric tube and request an X-ray; keep upright with suction and refer to paediatric surgery

E is right: polyhydramnios, frothy drooling and choking with feeds suggest oesophageal atresia (often with tracheo-oesophageal fistula). A tube that will not pass and coils in the upper pouch on X-ray confirms it; stop feeds, suction the pouch, nurse head-up and refer. A, B and C are wrong: further feeding risks aspiration pneumonia. D is wrong: antibiotics may be needed for aspiration but do not address the obstruction.

8. B · Diagnose a posterior cleft palate; support feeding with expressed breast milk by cup/spoon and refer to a cleft surgery service

B is right: a palpable notch with nasal regurgitation is a posterior (or submucous) cleft palate, which inspection alone misses; UCG advises helping the mother breastfeed or teaching an alternative method (cup feeding of EBM) and referral (e.g. CoRSU, Mulago). A is wrong: inspection alone is insufficient. C is wrong: standard teats are ineffective with a cleft; special feeding methods are needed. D is wrong: there is no thrush. E is wrong: breast milk should continue, and permanent NGT feeding is not needed.

9. D · Stop feeds, pass an NGT on free drainage, start IV fluids, inspect the anus, and refer urgently to paediatric surgery

D is right: bilious vomiting is a surgical emergency (malrotation with volvulus, atresia, Hirschsprung disease) until proven otherwise; with delayed meconium and distension, decompress, give IV fluids, check anal patency and position, and refer urgently. A and C are wrong: they treat constipation and delay surgery. B is wrong: almost all term babies pass meconium by 48 h, and bilious vomiting is never normal. E is wrong: soap enemas are dangerous.

10. A · Fractured right clavicle

A is right: crepitus and swelling over the clavicle with reduced arm movement (pseudoparalysis) after shoulder dystocia is a clavicle fracture; it heals in 7-10 days with gentle handling. B is wrong: CP is not diagnosed at birth and does not cause local crepitus. C is wrong: HIE causes generalised changes, not a unilateral finding with crepitus. D is wrong: Klumpke palsy affects the hand (absent grasp). E is wrong: septic arthritis appears days later with fever and swelling of the joint.

11. B · After the baby is at least 24 hours old, feeding well, with BCG and OPV0 given and danger-sign counselling done

B is right: UCG says do not discharge a baby under 24 hours old, assess breastfeeding before discharge, give BCG and polio 0, and counsel on danger signs and the 3-7-day visit. A is wrong: it is earlier than UCG advises. C, D and E are wrong: healthy term babies do not need to stay that long; regaining birth weight takes 10-14 days and the cord usually separates in 5-15 days.

12. D · Plantar grasp reflex

D is right: the plantar grasp persists until about 9-12 months (it must fade before walking). A is wrong: stepping fades by about 2 months. B is wrong: rooting fades by about 4 months. C is wrong: Moro fades by about 4-6 months. E is wrong: palmar grasp fades by about 3-6 months.

13. A · Avoid assigning sex, explain that tests are needed, check glucose and electrolytes, and refer urgently for possible congenital adrenal hyperplasia

A is right: ambiguous genitalia with impalpable gonads in an apparently female baby is most often congenital adrenal hyperplasia in a 46,XX baby; salt-wasting crises (hyponatraemia, hyperkalaemia, hypoglycaemia, shock) can occur in the first 1-2 weeks. Do not assign sex until assessed. B and C are wrong: guessing sex causes harm. D is wrong: delay risks an adrenal crisis. E is wrong: early surgery is not indicated.

14. E · Jaundice visible at 16 hours of age

E is right: jaundice in the first 24 hours is always pathological (haemolysis, sepsis); measure bilirubin urgently and treat. A, B, C and D are wrong: milia, physiological vaginal discharge, breast engorgement from maternal hormones and erythema toxicum are benign; reassure the mother.

15. C · Arrange a spinal ultrasound and neurosurgical review for possible occult spinal dysraphism

C is right: a deep dimple above the natal cleft with a hair tuft is an atypical marker of occult spinal dysraphism (tethered cord), which should be imaged (ultrasound in infancy) and reviewed. A is wrong: small, shallow dimples within the natal cleft are usually benign, but this one has high-risk features. B is wrong: probing can introduce infection into a sinus tract. D is wrong: cosmetic measures miss the underlying lesion. E is wrong: there is no infection.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The routine discharge check that wasn't routine

You are the junior clerk on the Mengo postnatal ward. A 22-year-old primigravida delivered a 3.0 kg girl by assisted breech delivery 22 hours ago at 39 weeks. The mother is HIV-negative (retested in labour), RPR negative, blood group A positive. The midwife asks you to do the discharge examination because the family wants to go home.

1. What will you check in the records and ask the mother before examining?

> Gestation, **breech presentation** (DDH and birth injury risk), labour/ROM duration, maternal fever, Apgar and resuscitation, **birth weight**, vitamin K/eye ointment/BCG/OPV0 given, maternal HIV/syphilis/HBsAg; ask about **feeding** (number of feeds, attachment), **urine and meconium**, and **any concerns**.

2. Describe the order you will use for the examination.

> Warm room, hands washed. **Observe** colour, breathing (RR 60 s), posture, movements, cry. **Measure** weight, OFC, length, temperature and plot. **Head-to-toe**: fontanelles and sutures, face, eyes (**red reflex**), ears, nose, **palate (look and feel)**, neck and clavicles, chest, heart (**femorals**, SpO2), abdomen and cord, genitalia, **anus**, limbs and feet. **Back and spine** in ventral suspension. Skin. **Neuro and reflexes**. **Hips last** (Barlow/Ortolani). Observe a breastfeed.

3. The left hip gives a clunk on abduction and lifting of the trochanter; abduction on that side is limited. What is this and what will you do?

> A **positive Ortolani test**: a dislocated, reducible left hip (**DDH**), with risk factors **breech, female, first-born**. Explain to the parents; avoid tight swaddling with legs extended; refer for **hip ultrasound and orthopaedic review** (Pavlik harness if confirmed); document carefully. Do not repeat the test forcefully.

4. She also has an asymmetric Moro (right arm moves less) and holds the right arm internally rotated with the wrist flexed, grasp intact. Interpretation?

> **Right Erb's palsy (C5-C6)** from traction during the breech delivery; exclude a clavicle/humerus fracture (palpate; X-ray if crepitus). Gentle handling, pin the sleeve for comfort in the first week, then **passive physiotherapy**; review regularly; refer if no biceps recovery by **3 months**.

5. Can she go home today? What will you tell the mother?

> She can go home after **24 h** if feeding well and the rest of the exam is normal, with **orthopaedic and physiotherapy appointments** arranged. Counsel: danger signs (not feeding, fever, fast breathing, convulsions, yellow palms/soles), cord care with chlorhexidine, warmth, exclusive breastfeeding, how to handle the arm and avoid tight swaddling, **3-7-day review** and **6-week immunisation and hip/arm check**.

Take-away: Risk factors from the history (breech, female, first-born) direct the exam: the hips and brachial plexus must be examined carefully before every discharge, and findings must lead to clear referral and follow-up.

Case 2: A baby with no known dates

A 17-year-old girl arrives at a HC IV in Hoima fully dilated and delivers a baby boy weighing 1.9 kg 20 minutes later. She has had no ANC and does not know her LNMP. The baby cries at birth. At 14 hours he is in skin-to-skin contact; temperature 36.4 C, RR 58, feeding 'a little'.

1. How will you estimate his gestation?

> **New Ballard score** (best at 12-20 h): 6 neuromuscular items (posture, square window, arm recoil, popliteal angle, scarf sign, heel-to-ear) and 6 physical items (skin, lanugo, plantar creases, breast, ear/eye, genitals). GA = (2 × score + 120) ÷ 5.

2. His Ballard score is 25. Interpret this together with his weight.

> (2 × 25 + 120) ÷ 5 = **34 weeks**. 1.9 kg at 34 weeks is about the **10th-50th centile → preterm AGA, LBW**. He is at risk of hypothermia, hypoglycaemia, poor feeding, jaundice and infection.

3. What else must you examine and check?

> Full head-to-toe exam; **glucose** (LBW, poor feeding); temperature (36.4 = cold stress); feeding observation; signs of infection; jaundice; vitamin K (0.5 mg preterm/LBW per local protocol), eye ointment, chlorhexidine; **mother's HIV, syphilis and HBsAg tests now** (no ANC), and start prophylaxis if positive.

4. What is your management plan?

> **Continuous KMC**, cover head; support breastfeeding, EBM by **cup** if suck weak (day 1 about 60 mL/kg/day ≈ 115 mL/day, 3-hourly ≈ 14 mL); glucose monitoring before feeds; daily weight; BCG and OPV0 before discharge; keep until feeding well, gaining weight 3 days and temperature stable 3 days (UCG small-baby care). Adolescent mother support, family planning.

5. On day 4 he is jaundiced to the lower abdomen and feeding 8 times a day by breast and cup. What do you do, and when can he go home?

> Measure **bilirubin** (TcB/serum) and plot it on a **preterm (34-week) threshold** (WHO: lower thresholds for babies below 35 weeks or with risk factors; NICE gestation-specific graphs); start **phototherapy** if above the line and keep feeding. Discharge (UCG small baby) when **breastfeeding well, gaining weight on 3 consecutive days, temperature normal for 3 days in KMC**, the mother is confident, and BCG/OPV0 are given; then weekly weighing until 3 kg.

Take-away: When dates are unknown, the New Ballard score plus weight lets you classify the baby (preterm vs SGA) and plan the right care.

Case 3: The 6-week check of an HIV-exposed infant

A 6-week-old girl born at term (birth weight 3.1 kg) attends Mengo MCH clinic with her mother, who is HIV-positive on TLD (viral load suppressed at 36 weeks). The baby has taken daily NVP since birth. Today she weighs 4.4 kg and length is 55 cm. The mother has no concerns.

1. What will you assess in the history at this visit?

> Feeding (**exclusive breastfeeding**, frequency, any other fluids), urine/stool, **NVP adherence** and any missed doses, mother's ART adherence and latest **viral load**, illnesses since birth, development (**social smile** by 6 weeks, follows faces), immunisations (BCG scar, OPV0), mother's health, family planning.

2. Interpret her growth.

> Weight gain 1.3 kg in 6 weeks ≈ **31 g/day** (normal about 25-35 g/day); plot weight-for-age, length-for-age and weight-for-length on **WHO charts**; expect Z-scores between -2 and +2.

3. Which parts of the newborn examination must be repeated at 6 weeks, and why?

> **Hips** (DDH can appear later), **red reflex** (cataract, retinoblastoma), **heart** (murmurs appear as pulmonary pressure falls; femoral pulses), **testes** (in boys), **fontanelle and head circumference** (hydrocephalus/microcephaly), **tone and reflexes** (early CP signs), **eyes fixing and following**, BCG scar.

4. What tests and treatments are due today?

> **DNA PCR** (HIV early infant diagnosis at 6 weeks), **cotrimoxazole prophylaxis** from 6 weeks until HIV is excluded per national guidelines, review NVP plan (low-risk: stop at 6 weeks; high-risk: continue NVP to 12 weeks per Uganda guidelines), **6-week vaccines** (OPV1, DPT-HepB-Hib1, PCV1, rotavirus 1 per UNEPI).

5. What counselling will you give?

> Continue **exclusive breastfeeding to 6 months**; mother's ART adherence and VL monitoring; return for PCR result; danger signs; next immunisation at **10 weeks**; family planning.

Take-away: The 6-week visit repeats the key newborn screens (hips, eyes, heart, growth, development) and adds HIV early infant diagnosis, cotrimoxazole and the first UNEPI vaccines for an HIV-exposed infant.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.